

State of Developer Ecosystem 2024: Language Trends, Satisfaction

Key insights from JetBrains' survey of 23,262 developers: language usage shifts (2017–2024), satisfaction, adoption intent, and domain spe

23,262

Survey respondents

2017–2024

Usage trend window

Top-10

Satisfaction ranking

What you'll get

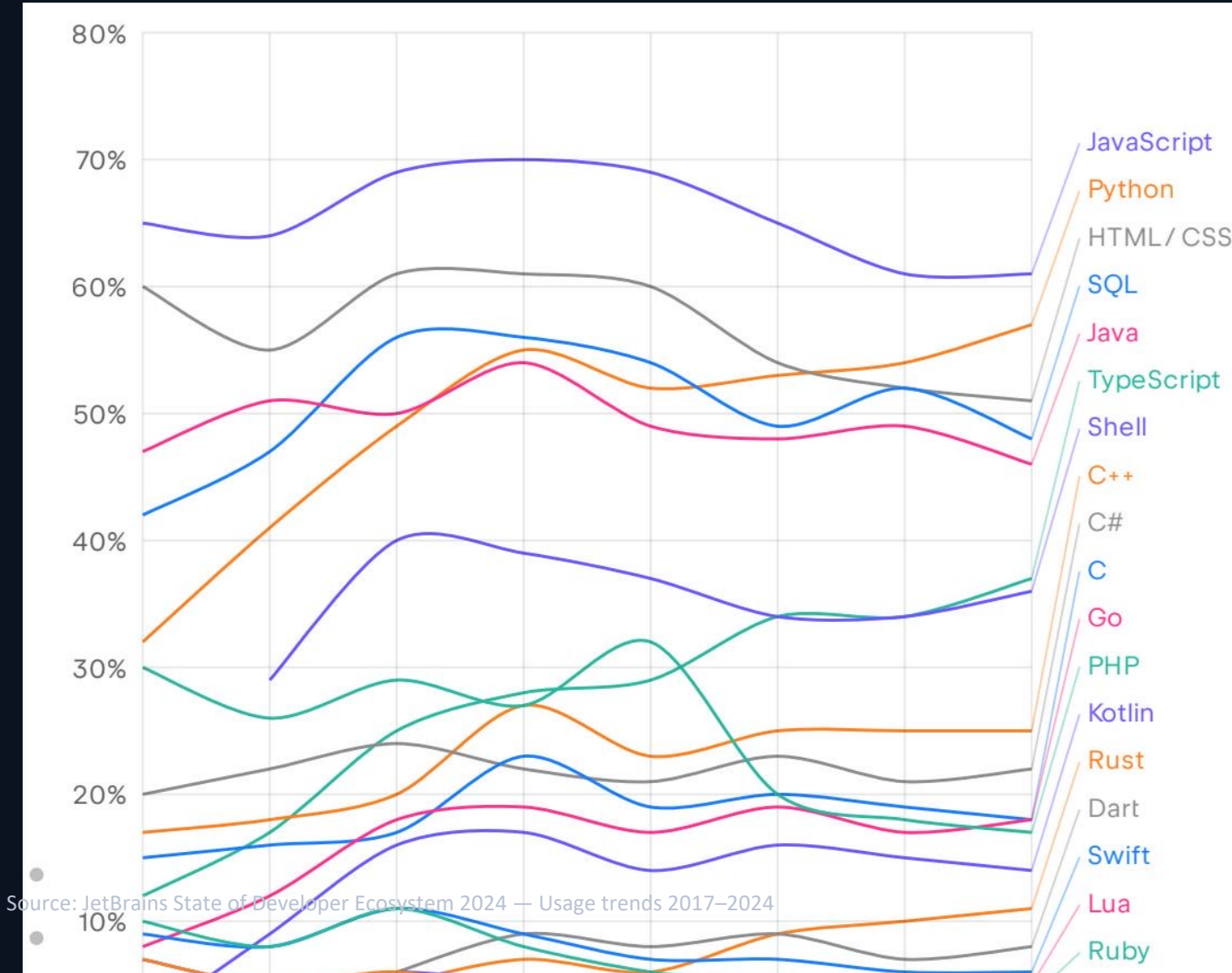
Where language usage is rising vs. declining

Which languages developers love most (and why that matters for retention)

How language choice maps to application domains (AI, UI, APIs, systems)

Python Closes the Gap: 8 Years of Language Usage Shifts

Overall usage share across the developer population (selected languages).



Signals for leaders

JavaScript remains #1 overall (61% in 2024) but declined from 65% (2017).

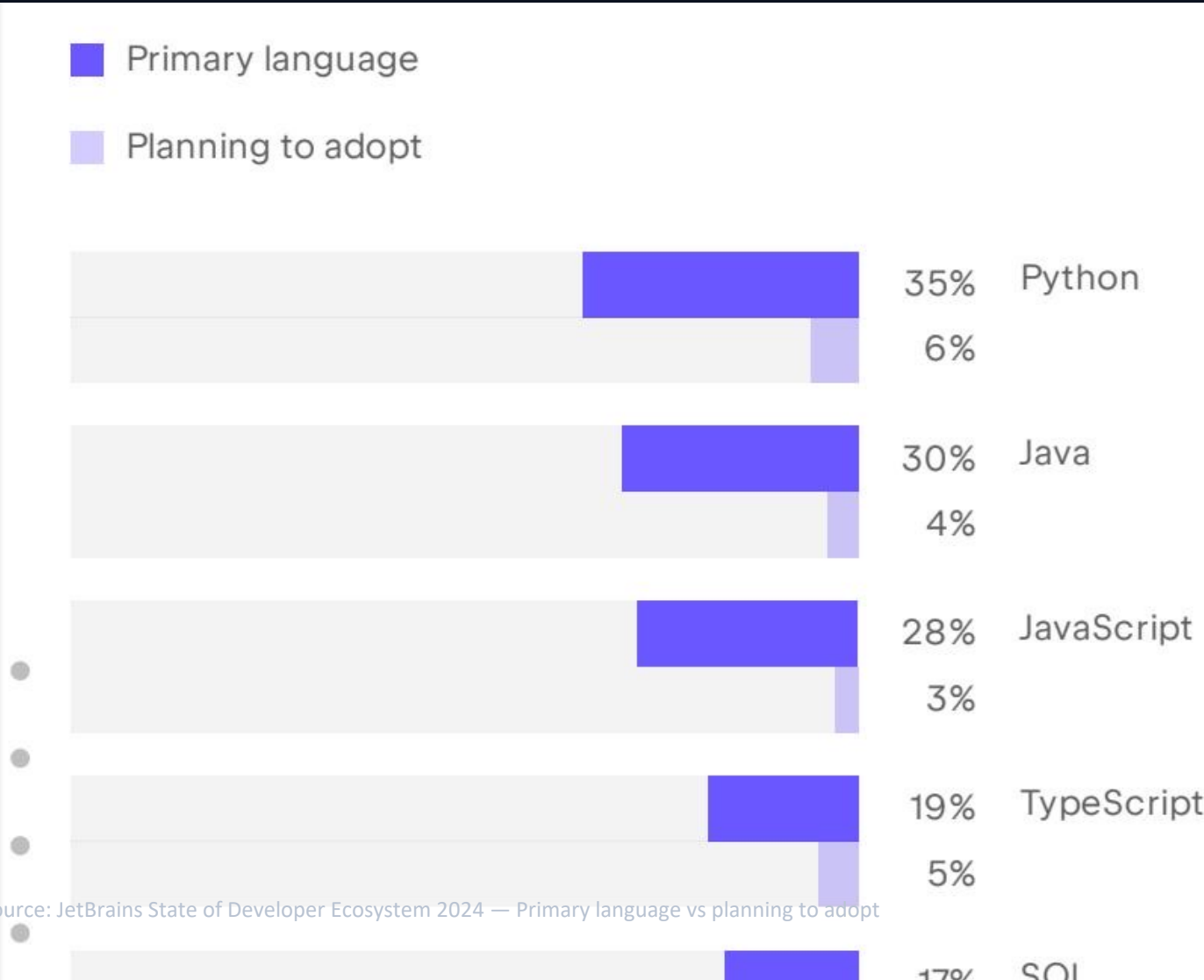
Python rose from 32% (2017) to 57% (2024), nearly matching JavaScript.

TypeScript tripled from 12% (2017) to 37% (2024).

Legacy web stacks continue sliding (PHP 30% → 17% from 2017).

Python Is Now the #1 Primary Language at 35%

Primary language indicates where teams spend most of their time; adoption intent shows willingness to switch.



Implications

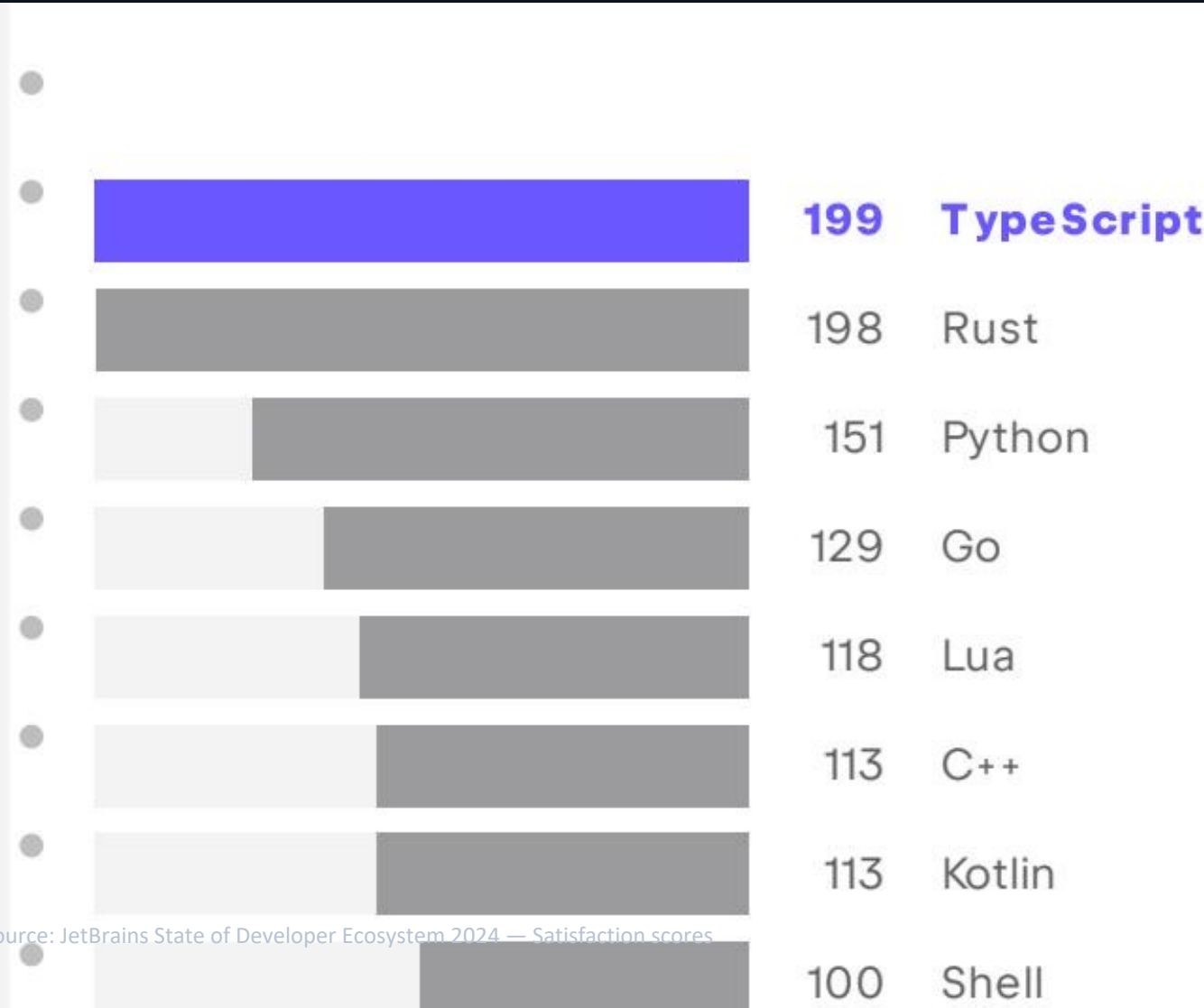
Primary stack changes are rare: adoption intent stays below 6%

Plan hiring and training assuming ecosystem lock-in and long-l

Use adoption intent as an early signal (Python 6%, TypeScript

TypeScript and Rust Top Developer Satisfaction, Nearly Tied at 1

Satisfaction score ranking (top 10). Higher is better.



What's driving sentiment

Type-safe, modern-tooling languages lead (TypeScript 199; Rust 198). Python remains highly liked (151) alongside strong ecosystem. Go (129) stays favored for pragmatic infrastructure work.

Each Language Owns a Domain: AI Belongs to Python, UI to TypeScript

Top language by application area (selected domains).

	Python	Java	JavaScript	TypeScript	SQL	C#	C++	PHP	
	38%	48%	38%	47%	25%	55%	17%	5%	Integrating with APIs and services
	35%	55%	38%	47%	37%	63%	34%	5%	Application logic and workflows
	35%	17%	7%	6%	45%	18%	14%	7%	Data processing and analytics
	31%	5%	3%	4%	6%	6%	11%	1%	Artificial intelligence and machine learning
	28%	49%	22%	29%	31%	50%	13%	5%	Providing APIs and services
	27%	5%	3%	2%	21%	5%	6%	1%	Data science / Data analytics
	26%	9%	10%	12%	5%	12%	9%	1%	Integration with AI
•	22%	4%	9%	7%	11%	8%	4%	1%	Ad hoc scripts
•	19%	29%	23%	31%	11%	36%	30%	3%	Libraries / Frameworks
•	16%	13%	9%	15%	4%	18%	9%	1%	Automating software testing and quality assurance
•									
•	14%	19%	58%	62%	8%	32%	18%	2%	User interface (handling user inputs and interactive elements)
•									
•									

Source: JetBrains State of Developer Ecosystem 2024 — Language use cases by domain

Domain leaders (from survey)

AI / Machine Learning: Python (31%)

Data science / analytics: Python (28%); Data processing & analytics: Python (35%)

User interface: TypeScript (62%) and JavaScript (58%)

APIs & services: Java (49%); Application logic & workflows: Java (55%)

Systems tools & components: C++ (27%)

What This Means for Your Technology Strategy in 2025

Translate trends and sentiment into hiring and platform priorities.

Hiring priorities

Scale Python talent for AI/ML and data roles (Python leads AI/ML at 31%)

Treat TypeScript as a strategic frontend/full-stack default (UI: 62%)

Keep Java strong for services and workflow-heavy backends (APIs: 49%; logic: 58%)

Use C++ depth where systems and performance dominate (systems tools: 27%)

Platform / DX actions

Invest in type safety and developer tooling: satisfaction leaders are TypeScript (62%)

Assume ecosystem lock-in: adoption intent stays under 6% for every language

Plan exits for shrinking stacks where appropriate (e.g., PHP down to 17% overall)